

Miracle Advance Technologies

DIGITAL EXPERIENCE HEALTH CHECK Vietnam

Application User Manual

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1.1. Organization of the Manual

The manual is designed to initially familiarize you with the application's most common features, such as dashboards and the search feature. While doing so, the most frequent elements are explained so that you can later relate to the terms as you explore the application's features and functionalities in greater depth. Items in the Main Menu will vary depending on user responsibilities, but the complete menu is studied as the manual progress, and user roles are explained ahead of time, so the menu options that each user position sees may be understood. The Study Worklist includes the most fundamental and necessary patient details that the doctors perform. Best efforts were made to write the features in a way that follows the chronology used by the majority of user roles in the system. This section delves deeply into the hospital's multiple operations and management of inpatients. The documentation advances in this part based on the submenu within the Digital Experience Health Check application. Following that, the sections that follow will highlight the various tools and capabilities within the application that you may find more useful once you've gotten to know your way around Digital Experience Health Check.

1.2. Additional Help

Throughout the user manual, common terms are bolded to familiarize the reader with the terms used in the application. Section numbers are indexed (numbered) and/or linked next to popular keywords and task phrases, allowing the reader to go directly to the section for more

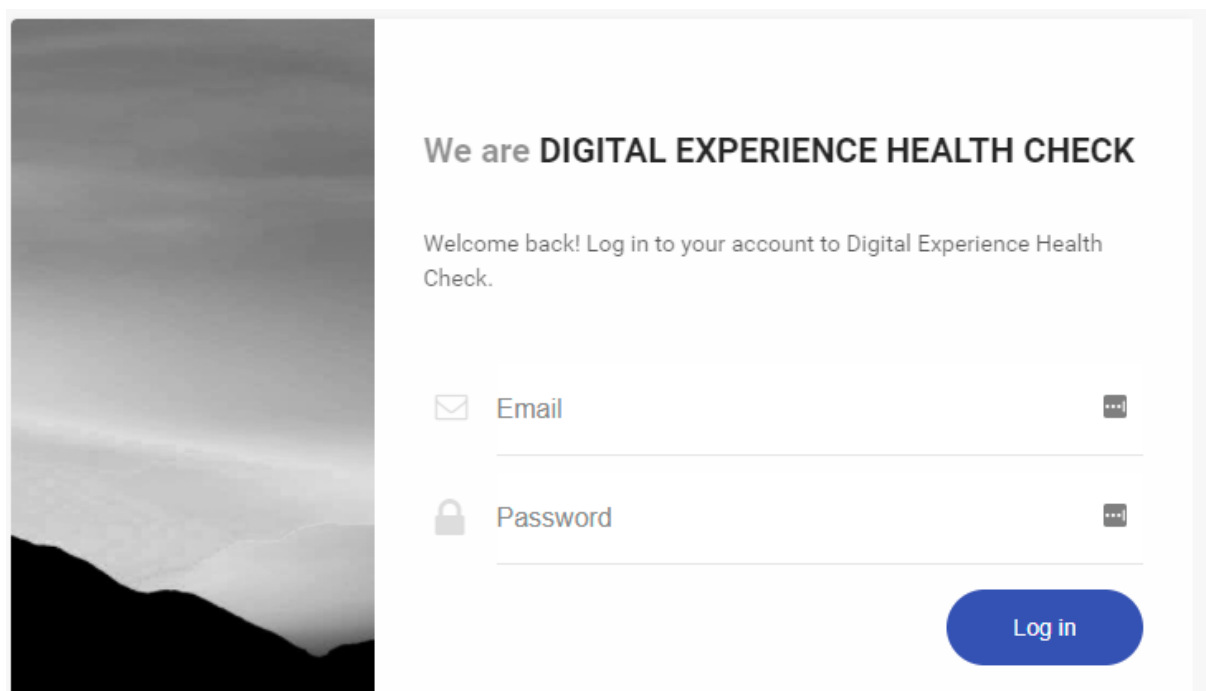
information as needed. Keywords in blue are linked for easier navigation, so you may return to the part where you can get specifics about keywords/phrases.

1.3. Browser Requirements

- Latest Chrome browser
- Edge
- Latest Mozilla Firefox
- Internet Explorer
- Safari

1.4. Getting Started

Before you can log in to Digital Experience Health Check, you will need to have your email registered by the System Administrator to be a user of the system. Go to [link](#) and log in with your credentials. The login page should look like this



We are **DIGITAL EXPERIENCE HEALTH CHECK**

Welcome back! Log in to your account to Digital Experience Health Check.

Email

Password

Log in

Once logged in, you should be viewing the [Study Worklist](#). From here you will find the data of this patient.

1.5. Study Worklist

From this Study Worklist, the user will find different options to operate the data.

Study Worklist : Data Center

From Date : 03-12-2021 To Date : 03-12-2021 Hospital : 92 items selected Group : 137 items selected Search Today Yesterday Last 3 Day Last 7 Day

Show All Auto Refresh Selected Study(s) : 0 / 161

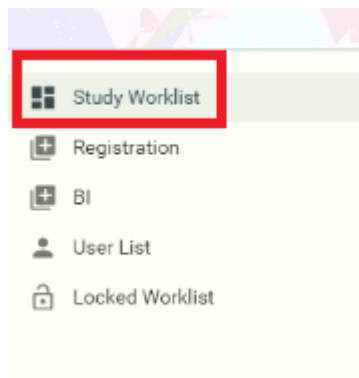
Import Studies Add to PACS Queue Appointment List Appointment Card Finalized Report Scan Accession#

☐ Completed ☐ Finalized ☒ Completed & Finalized ☐ Scheduled/In-progress

14 columns selected CSV Export : All CSV Export : Selection Only

☐	Queue#	Access	HN	Patient Name	Date of Birth	Gender	Exam Name	Image	Status	Finalized Category	Compared	Created On	Abnormality	Lung Opacity	Action
☐	✓	21120	ISO5	รศ.ดร.วิมล วิมล	08-11-2021	Female	CHEST PA UPRIGHT	1	Finalized	Category 1	Stable	03-12-2021 08:59 AM	0.1 %	0.1 %	       

In the left corner, the user will find the **Study Worklist**.



1.6. Understanding the Features

Before you can start to use Digital Experience Health Check, it is imperative that you learn about the small features that are incorporated throughout the application for ease of use, convenience, and better user experience.

The most default feature you will find in almost any tabulated information list is the Column Settings option. Here's how you see it:

Study Worklist : Data Center

From Date : 14-12-2021 To Date : 14-12-2021 Hospital : Miracle Group : CHD RAY HOSPITAL Search Today Yesterday Last 3 Day Last 7 Day

Show All Auto Refresh Selected Study(s) : 0 / 0

Import Studies Finalized Report Summary Report Scan Accession#

☒ Completed ☐ Finalized ☐ Completed & Finalized ☐ Scheduled/In-progress

13 columns selected CSV Export : All CSV Export : Selection Only

☐	Case No	Accession#	HN	Patient Name	Date of Birth	Gender	Exam Name	Image	Status	Created On	Abnormality	Print Count	Lung Opacity	Action
--------------------------	---------	------------	----	--------------	---------------	--------	-----------	-------	--------	------------	-------------	-------------	--------------	--------

10

HN	Patient Name	Gender	Exam Name	Image	Status	Finalized Category	Compared	Created On	Abnormality	Lung Opacity	Technical Quality	Action
05	IS0523571	ภาณุพงศ์	Female	CHEST PA UPRIGHT	1	Finalized	Category 1	Stable	03-12-2021 08:59 AM	0.1 %	0.1 %	A

Hospital: Here the user can select items from the dropdown.

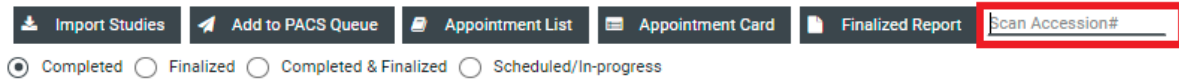
Group: Here the user can select group items from the dropdown.

Show All: The user can find all types of status by simply clicking to Show all toggle button in the right corner of the application.

Queue#	Accession#	HN	Patient Name	Gender	Exam Name	Image	Status	Finalized Category	Compared	Created On	Abnormality	Lung Opacity	Technical Quality	Action
50	20211203CI	22020/64	NIRAMON CHATHIRAN	Female	CHEST PA PORTABLE	1	Finalized	Category 1	Stable	03-12-2021 08:49 AM	66 %	66 %	A	

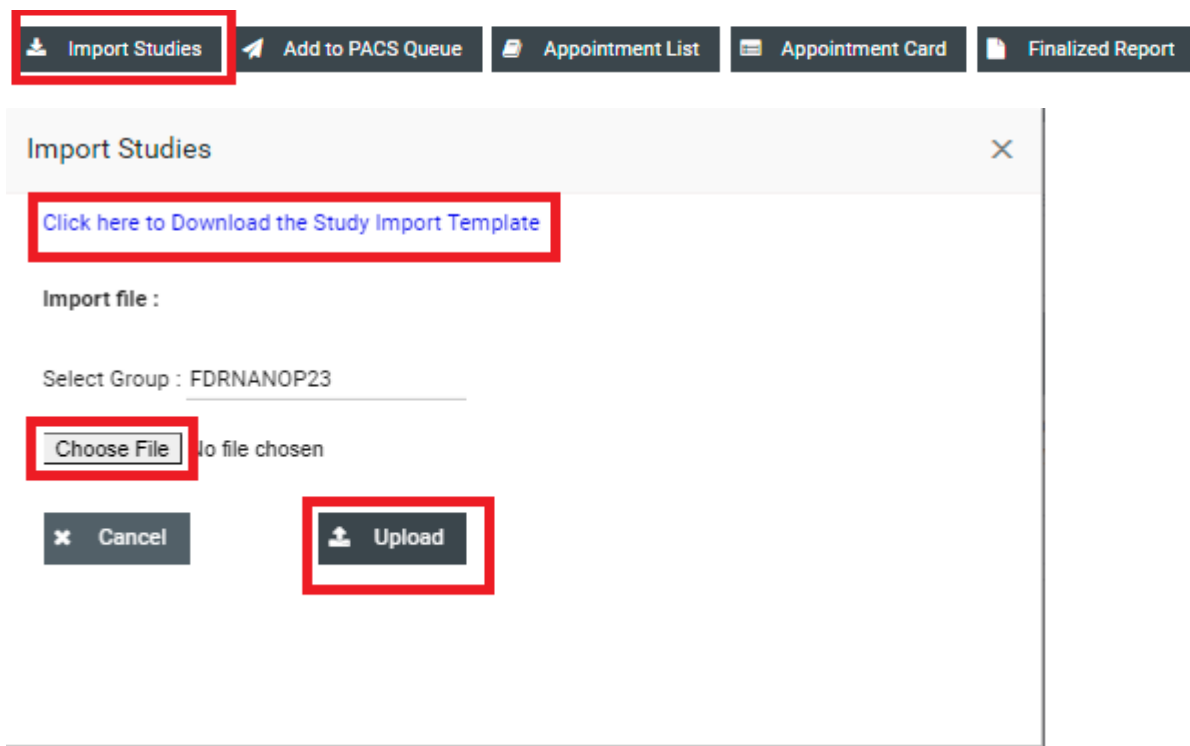
Scan Accession:

The user can scan the accession right after the registration for showing the arrival schedule of the patient.



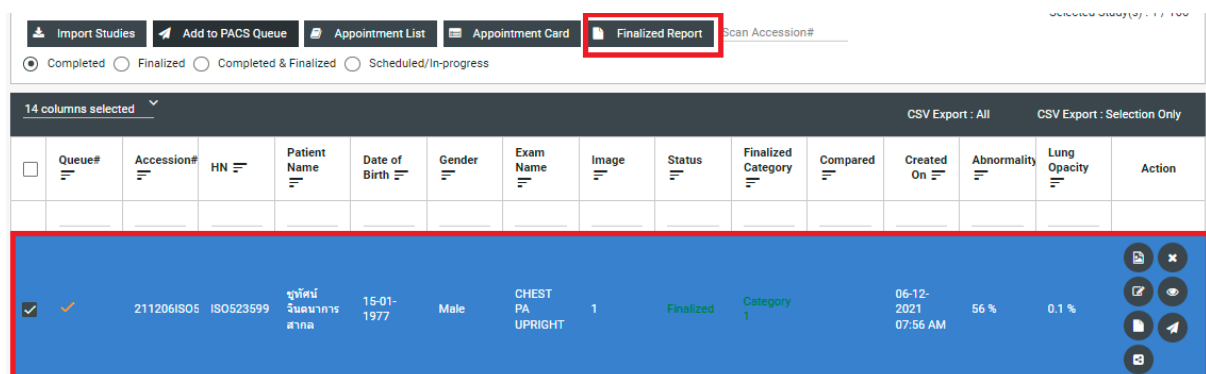
Import Studies

The user can import studies and also download the study Import templates.



Finalized Report

The user can see the finalized report by selecting any finalized status.



FINALIZED REPORT

First Name : ชูทัศน์	Study : CHEST PA UPRIGHT
Last Name : ชื่นชมการวิสาคร	SSN : 3909800050678
Age : 44.00	HN : ISO523599
Gender : Male	Finalized By : RCRT Team
	Created On : 06.12.2021

Findings :

Lungs : Normal

Pleura : Normal

Heart and Mediastinum: Normal

Others : Normal

Impression :

Category; Category 1 : Normal chest X-ray or no abnormality detected.

DISCLAIMER : Abnormality Category adapted from Rama Co-RAIS version 1

AI Findings :

Abnormality Score : 56 %

AI Finding Report :

Tuberculosis : 56%

Atelectasis : 0.1%

Cardiomegaly : 0.1%

Pulmonary Edema : 0.1%

Lung Opacity : 0.1%

Nodule : 0.1%

Mass : 0.1%

Pleural Effusion : 0.1%

Summary Report

The user can see the summary report by selecting any summary status.

Status

The user can select status individually to see the specific column.

☐ Completed
 ☒ Finalized
 ☐ Completed & Finalized
 ☐ Scheduled/in-progress

14 columns selected														CSV Export : All	CSV Export : Selection Only
	Queue#	Accession#	HN	Patient Name	Date of Birth	Gender	Exam Name	Image	Status	Finalized Category	Compared	Created On	Abnormality	Lung Opacity	Action
☐	✓	211206ISO5	ISO523599	ชูทัศน์ ชื่นชมการวิ สาคร	15-01- 1977	Male	CHEST PA UPRIGHT	1	Finalized	Category 1		06-12- 2021 07:56 AM	56 %	0.1 %	

On the right side of the column, the user will find the Action column.

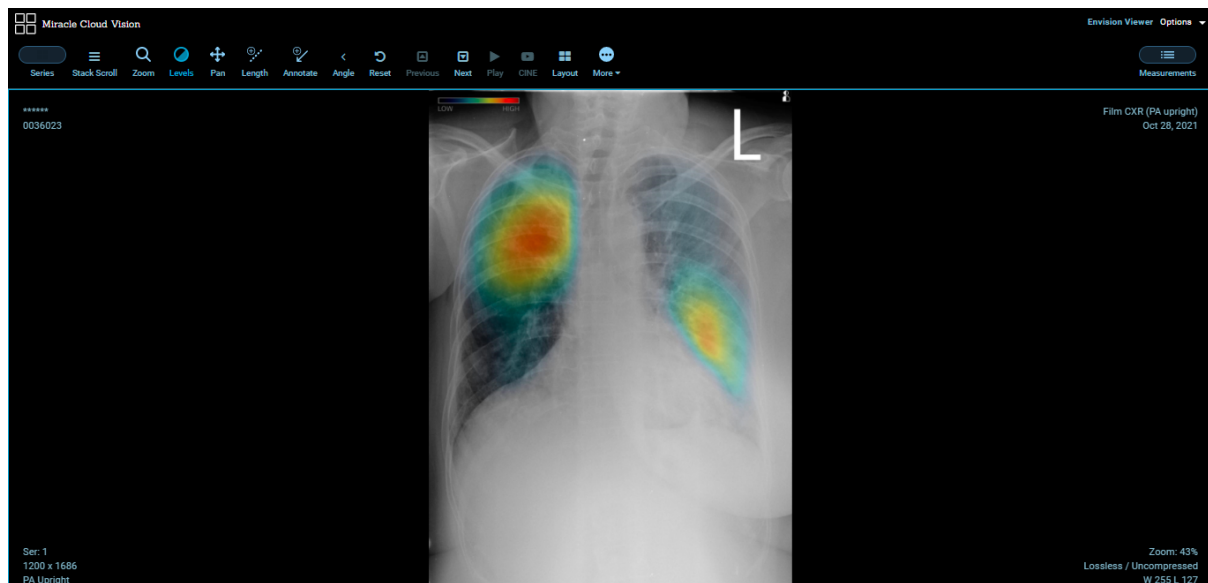
14 columns selected														CSV Export : All		CSV Export : Selection Only
	Queue#	Accession#	HN	Patient Name	Date of Birth	Gender	Exam Name	Image	Status	Finalized Category	Compared	Created On	Abnormality	Lung Opacity	Action	
☐	☒	211206IS05	IS0523599	สุวิทย์ วิเศษธรรม สกล	15-01- 1977	Male	CHEST PA UPRIGHT	1	Finalized	Category 1		06-12- 2021 07:56 AM	56 %	0.1 %		

The user will find some buttons in the Action column.

- **View AI Image**
- **Cancel**
- **Reconcile**
- **View Result**
- **Copy Shareable Link**

View AI Image

The user can see the AI image by clicking the **View AI Image**.



This is a DICOM image, where the AI marks the abnormal area and gives the output.

The user will find the AI findings by clicking the **View Result**.

Cancel

This action helps to remove the study from the existing column. After clicking the button,

 the user will get a confirmation pop-up. Also, you have to provide a reason for canceling the study.

Confirmation

i Are you sure to remove this study?

Reason : Because of some confusion.

X No ✓ Yes

Reconcile

The user can update study info by clicking **Reconcile** button.

Update Study Info

Patient Information

SSN : 123
HN : 5399378
First Name : Phumiphat
Last Name : Photichanprasert
DOB : 19-04-1989
Gender : Male
Age : 32

Study Information

Queue Serial : 4
Accession No : 180430SDOCR431
Study Date :* 02-11-2021
Study Name :* Chest

Confirm Confirm & Generate MWL

Reporting

First Name : นางสาวณัฏฐาณันท์ Last Name : Age : 55 Gender : Female
Accession No : 2470419933 Study Name : Film CXR (PA upright) HN : 0036023 SSN :
Queue Serial : 2 Study Date : Dec 7, 2021 Image : 1 RT-PCR positive Date :
Comparison : X No DICOM Viewer : Open Counter Status : 0 : 78 : 130 Status : Completed

Technical Quality
☐ Good
☒ Acceptable with some technical defect, but still adequate for interpretation.
☐ Unacceptable for interpretation. Repeat chest X-ray is required.

Template:
Select Template

AI Finding

Abnormality Score : 56 %

AI Finding Report :

Tuberculosis : 56%

Atelectasis : 0.1%

Cardiomegaly : 0.1%

Pulmonary Edema : 0.1%

Lung Opacity : 0.1%

Nodule : 0.1%

Mass : 0.1%

Pleural Effusion : 0.1%

AI DISCLAIMER: This is an AI generated Finding.

DISCLAIMER:

When the user doubles click the column, they will find **Reporting** Page and **Envision DICOM Viewer**.

They will first see the patient details and can select the Technical Quality based on AI Findings. Moreover, they can select Template.

Reporting

First Name : นาง ดวงมณี แก้วสิงห์

Last Name :

Age : 55

Gender : Female

Accession No : 2470419933

Study Name : Film CXR (PA upright)

HN : 0036023

SSN :

Queue Serial : 2

Study Date : Dec 7, 2021

Image : 1

RT-PCR positive Date :

Comparison : ✕ No

DICOM Viewer : 🔍 Open

Counter Status : 0 : 78 : 130

Status : Completed

Technical Quality

☐ Good
 ☒ Acceptable with some technical defect, but still adequate for interpretation.
 ☐ Unacceptable for interpretation. Repeat chest X-ray is required.

Template:

Select Template 📄

In below, the user will find some options.

- AI/History
- Finding
- Impression

AI/History

Here, the user will find the AI Findings of the patient which are detected by the AI.

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Case No	Accession#	Exam Name	Exam Date	Actions
	180302DIACD001	Chest	04-11-2021 01:56 PM	 
TEMP001	180820SDOCR215	Chest	02-11-2021 06:10 PM	 
	190513SDOCR369	Chest	01-12-2021 04:17 PM	 

AI/History

Finding

Remarks

Patient History

Any significant abnormality?

No

Significant abnormalities

AI DISCLAIMER: This is an AI generated Finding.

DISCLAIMER: Categories and the corresponding levels of suspicion for pulmonary involvement in a patient with confirmed COVID-19 is based on the proposal (Version 26/04/2021) by the Division of Diagnostic Radiology, Department of Diagnostic and Therapeutic Radiology, Faculty of Medicine, Ramathibodi Hospital, Mahidol University.

Report :

Findings :

Can Suggest Tuberculosis : Normal

OTHER X-Ray Findings : Normal

Remarks :

NORMAL

Mark as Normal

Save as Final

Skip

Load Next Patient

From Patient History, the user can **View AI Images** and **Report**.

The user can select findings by own if needed. However, AI has already provided the findings.

Case No	Accession#	Exam Name	Exam Date	Actions
	180302DIACD001	Chest	04-11-2021 01:56 PM	 
TEMP001	180820SDOCR215	Chest	02-11-2021 06:10 PM	 
	190513SDOCR369	Chest	01-12-2021 04:17 PM	 

AI/History

Finding

Remarks

Any significant abnormality?

No

Significant abnormalities

AI DISCLAIMER: This is an AI generated Finding.

DISCLAIMER: Categories and the corresponding levels of suspicion for pulmonary involvement in a patient with confirmed COVID-19 is based on the proposal (Version 26/04/2021) by the Division of Diagnostic Radiology, Department of Diagnostic and Therapeutic Radiology, Faculty of Medicine, Ramathibodi Hospital, Mahidol University.

Report :

Findings :

Can Suggest Tuberculosis : Normal

OTHER X-Ray Findings : Normal

Remarks :

NORMAL

Mark as Normal

Save as Final

Skip

Load Next Patient

If the user clicks the check box of **Significant abnormalities** then the user will find some other abnormalities. They can select those abnormalities with the patient's conditions.

Case No	Accession#	Exam Name	Exam Date	Actions
	180302DIACD001	Chest	04-11-2021 01:56 PM	 
TEMP001	180820SDOCR215	Chest	02-11-2021 06:10 PM	 
	190513SDOCR369	Chest	01-12-2021 04:17 PM	 

AI/History

Finding

Remarks

Any significant abnormality?

No

Significant abnormalities

AI DISCLAIMER: This is an AI generated Finding.

DISCLAIMER: Categories and the corresponding levels of suspicion for pulmonary involvement in a patient with confirmed COVID-19 is based on the proposal (Version 26/04/2021) by the Division of Diagnostic Radiology, Department of Diagnostic and Therapeutic Radiology, Faculty of Medicine, Ramathibodi Hospital, Mahidol University.

Report :

Findings :

Can Suggest Tuberculosis : Normal

OTHER X-Ray Findings : Normal

Remarks :

NORMAL

Mark as Normal

Save as Final

Skip

Load Next Patient

AI/History	Finding	Remarks
☒ Any significant abnormality? ☒ Can Suggest Tuberculosis ☐ OTHER X-Ray Findings	Can Suggest Tuberculosis ☐ Infiltrate or consolidation ☐ Cavitory lesion ☐ Pleural effusion ☐ Miliary findings ☐ Discrete nodule(s) without calcification ☐ Other ☐ Reticular markings suggestive of fibrosis ☐ Nodule or mass with poorly defined margins 10 -(such as tuberculoma) ☐ Hilar/mediastinal adenopathy ☐ Discrete linear opacity ☐ Volume loss or retraction ☐ Irregular thick pleural reaction	

When the user selects any of the findings from **Can Suggest Tuberculosis**, the button **Save as Final** will appear.

AI/History	Finding	Remarks
☒ Any significant abnormality? ☒ Can Suggest Tuberculosis ☐ OTHER X-Ray Findings	Can Suggest Tuberculosis ☐ Infiltrate or consolidation ☒ Cavitory lesion ☐ Pleural effusion ☐ Miliary findings ☐ Discrete nodule(s) without calcification ☐ Other ☐ Reticular markings suggestive of fibrosis ☐ Nodule or mass with poorly defined margins 10 -(such as tuberculoma) ☐ Hilar/mediastinal adenopathy ☐ Discrete linear opacity ☐ Volume loss or retraction ☐ Irregular thick pleural reaction	Report : Findings : Can Suggest Tuberculosis: Cavitory lesion, OTHER X-Ray Findings : Normal Remarks : NORMAL <input checked="" type="button" value="Mark as Normal"/> <input checked="" type="button" value="Save as Final"/> Skip

Otherwise, the button becomes blurred.

AI/History	Finding	Remarks
☒ Any significant abnormality? ☒ Can Suggest Tuberculosis ☐ OTHER X-Ray Findings	Can Suggest Tuberculosis ☐ Infiltrate or consolidation ☐ Cavitory lesion ☐ Pleural effusion ☐ Miliary findings ☐ Discrete nodule(s) without calcification ☐ Other ☐ Reticular markings suggestive of fibrosis ☐ Nodule or mass with poorly defined margins 10 -(such as tuberculoma) ☐ Hilar/mediastinal adenopathy ☐ Discrete linear opacity ☐ Volume loss or retraction ☐ Irregular thick pleural reaction	Report : Findings : Can Suggest Tuberculosis : Normal OTHER X-Ray Findings : Normal Remarks : NORMAL <input checked="" type="button" value="Mark as Normal"/> Save as Final Skip

Remarks

In this section, the user can give remarks to the patient. This remark will be added to **Report**.

Reporting

☐ Good
☒ Acceptable with some technical defect, but still adequate for interpretation.
☐ Unacceptable for interpretation. Repeat chest X-ray is required.

AI/History Finding **Remarks**

The patient need to take rest

AI DISCLAIMER: This is an AI generated Finding.

DISCLAIMER: Categories and the corresponding levels of suspicion for pulmonary involvement in a patient with confirmed COVID-19 is based on the proposal (Version 26/04/2021) by the Division of Diagnostic Radiology, Department of Diagnostic and Therapeutic Radiology, Faculty of Medicine, Ramathibodi Hospital, Mahidol University.

Report :

Findings :

Can Suggest Tuberculosis : Normal
 OTHER X-Ray Findings : Normal

Remarks :

The patient need to take rest

NORMAL

☒ **Mark as Normal**
☐ Save as Final
☐ Skip
☒ Load Next Patient

2. Registration

In this section, the user can register for the study.

Study Registration

Patient Information

First Name		Last Name	
DOB		Gender	
Age		HN	☒ Auto Generate
SSN *	☒ Auto Generate		

Study Information

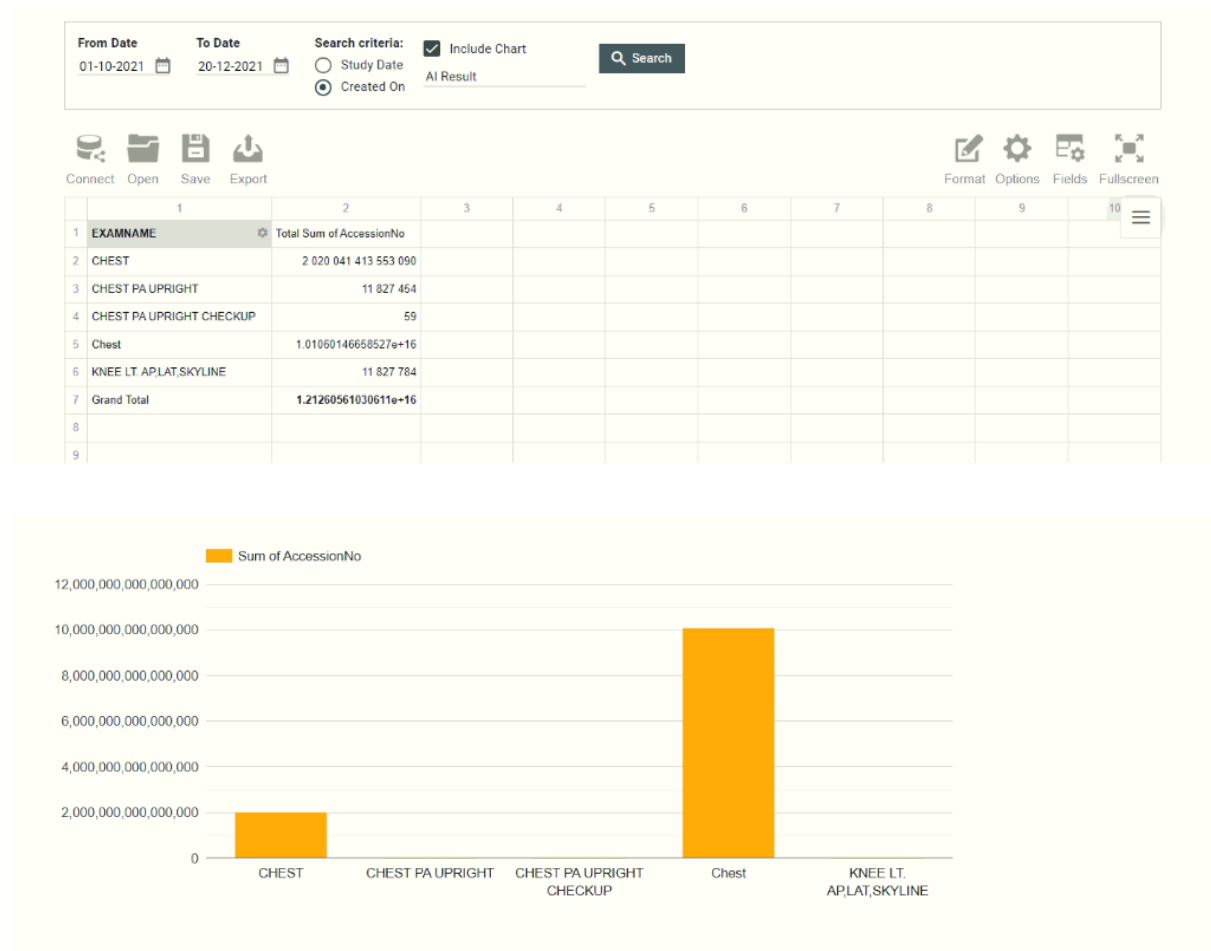
Accession No :	☒ Auto Generate	Selected Group *	CHO RAY HOSPITAL
Queue# *		Select Study : *	Chest

Note: * Mandatory Field

HN, SSN, and Accession No will be auto-generated when the user clicks the checkbox.

4. BI

In this Business Intelligence section, the user will get raw data and statistical data for better visualization.



In the first part, the user can search data, From Date -To Date. They can also select whether they want to see **Study Date** and **Created On** from the **Search Criteria**. Also, the user can select the **Include Chart** option to view the visualization of the data.

In the left corner users will find some functions like- Connect, Open, Save, and Export.

Connect

In this option, the user can connect any data from-

- To Local CSV
- To Local JSON
- To remote CSV
- To remote JSON

Business Intelligence

From Date: 07-12-2021 To Date: 07-12-2021 Search criteria: ☒ Include Chart ☐ Study Date ☒ Created On AI Result

Connect menu_connect Export Format Options Fields Fullscreen

To local CSV
To local JSON
To remote CSV
To remote JSON

	2	3	4	5	6	7	8	9	10
1	EXA	sum of AccessionNo							
5	Chest AP Portable	250 012 531 800 472							
6	Film CXR (PA upright)	7 411 250 818							
7	UNKNOWN	570 050 971 101 201							
8	Grand Total	8 668 948 161 060 576							

They can also open Local Report or Remote Report from the **Open** option.

Business Intelligence

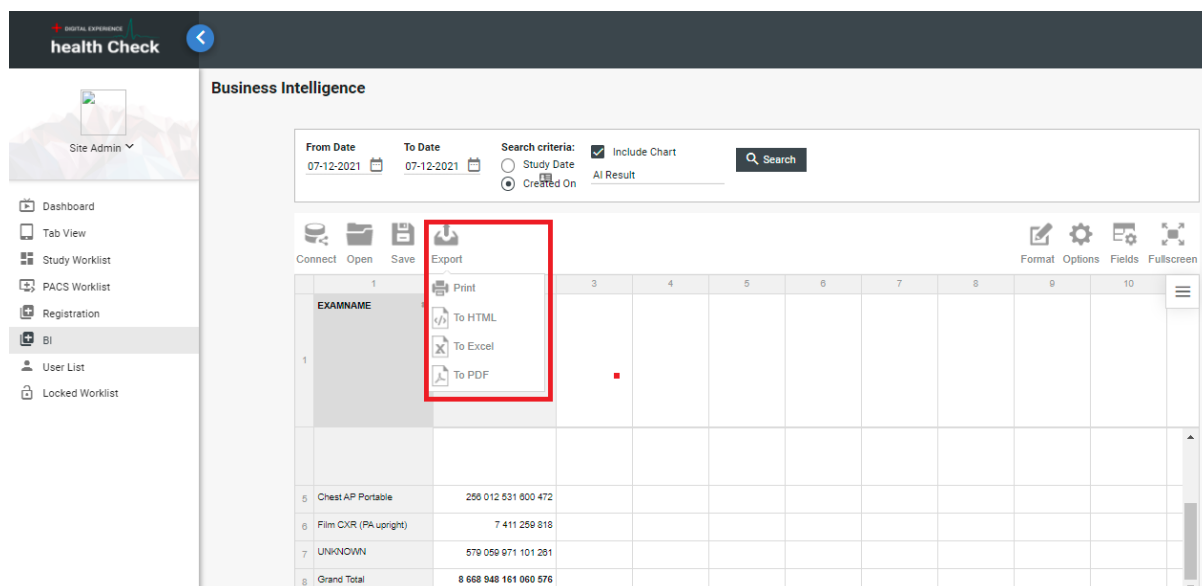
From Date: 07-12-2021 To Date: 07-12-2021 Search criteria: ☒ Include Chart ☐ Study Date ☒ Created On AI Result

Connect Open Save Export Format Options Fields Fullscreen

Local report
Remote report

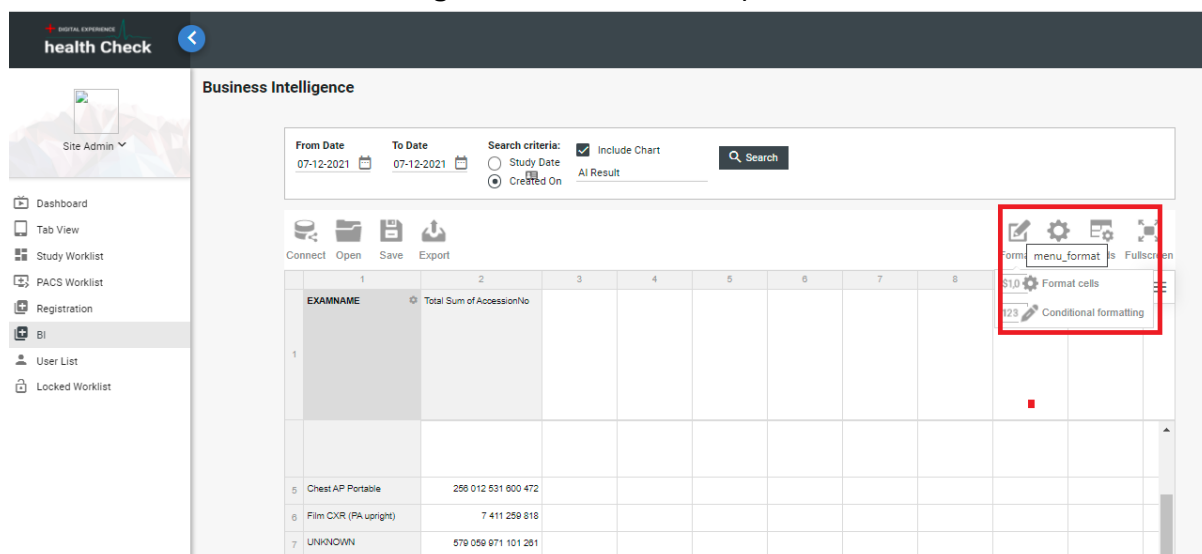
	2	3	4	5	6	7	8	9	10
1	EXA	sum of AccessionNo							
5	Chest AP Portable	250 012 531 800 472							
6	Film CXR (PA upright)	7 411 250 818							
7	UNKNOWN	570 050 971 101 201							
8	Grand Total	8 668 948 161 060 576							

The user can also **Print**, export to HTML, Excel, and PDF format by clicking on the **Export** option.



In the right corner, the user will find **Format**, **Options**, **Fields**, and **Full-Screen** functions.

The user can easily format the data by clicking on the **Format**. The user will find **Format cells** and **Conditional formatting** inside the **Format** dropdown.



In the **Format cells**, the user will find some basic format changes options.

Format cells

APPLY

CANCEL

CHOOSE VALUE

Choose value

Text align

right

Thousand separator

(Space)

Decimal separator

.

Decimal places

None

Currency symbol

Currency align

left

Null value

Format as percent

false

In **Conditional formatting**, the user will first find a pop-up window where they have to choose the ‘+’ button for further proceeding.

Conditional formatting

+

APPLY

CANCEL

There they can choose the format either less than, less than, or equal, and so on. Moreover, they can also choose a color for those format changes.

Conditional formatting

+

APPLY

CANCEL

Value:

All values

Less than

0

Format:

Arial

12px

A

73.93

The user can drag and drop fields to arrange the data inside the **Fields** option.

Fields

Add calculated value
APPLY
CANCEL

Drag and drop fields to arrange

All Fields
Expand All

> CreatedOn
> Dob
> OrderDate
☐ AbnormalityScore
☒ AccessionNo
☐ AtelectasisScore
☐ AtelectasisText
☐ CalcificationScore

Report Filters

Drop field here

Columns

Σ Values

Drop field here

Rows

ExamName

Drop field here

Values

Sum of AccessionNo Σ ∨

Drop field here

Note: The user can **Add calculated value** from the **Fields**.
Finally, they can maximize the screen with the **Fullscreen** option.

3. User List

You will find the details of User lists from here. Moreover, the user can update any users from the list. Also, they can add new users from the **User List**.

User List

+ Add New User

6 columns selected

Global filter

Full Name	Email	Mobile No	Created On	Status	Role Name	Action
Wariya Chintanapakdee	cwariya@hotmail.com	128		Approved	Radiologist	Update
Chomphunut Vijitsanguan	Chompoovj@gmail.com	818696369		Approved	Radiologist	Update
Wanwarang Teerasamit	Mojamp@gmail.com	0896986189		Approved	Radiologist	Update
Jitladda Wasinrat	jitladda_siriraj@hotmail.com	818265342		Approved	Radiologist	Update
Chalida Aphinives	achali@kku.ac.th	0857583636		Approved	Radiologist	Update

The user will get a form after clicking the **Add New User**.

User Information Entry

First Name: *

First Name

Last Name: *

Last Name

Phone No: *

Phone No

Email: *

Email

Password: *

Password

Confirm Password: *

Confirm Password

User Role: *

Select Role

Create

Create & Approve

Cancel

Note: * Mandatory Fields

4. Locked Worklist

The user will get the locked study from the **Study Worklist**.

Locked Worklist

Selected Group(s) :

135 items selected

Search

7 columns selected

Global filter

	Locked By	Locked On	Accession#	HN	Patient Name	Exam Name	Status	Action
☐								

1

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